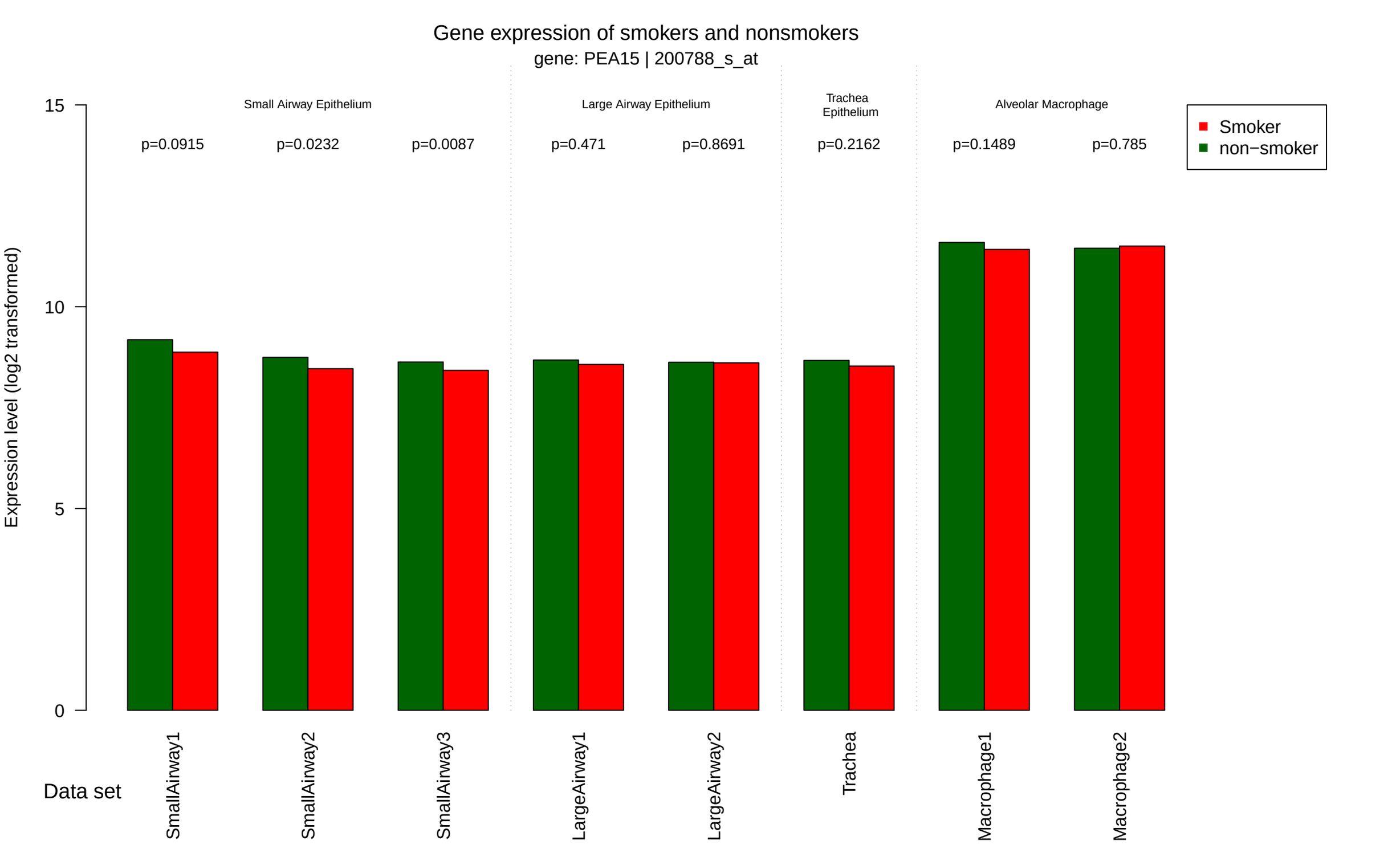
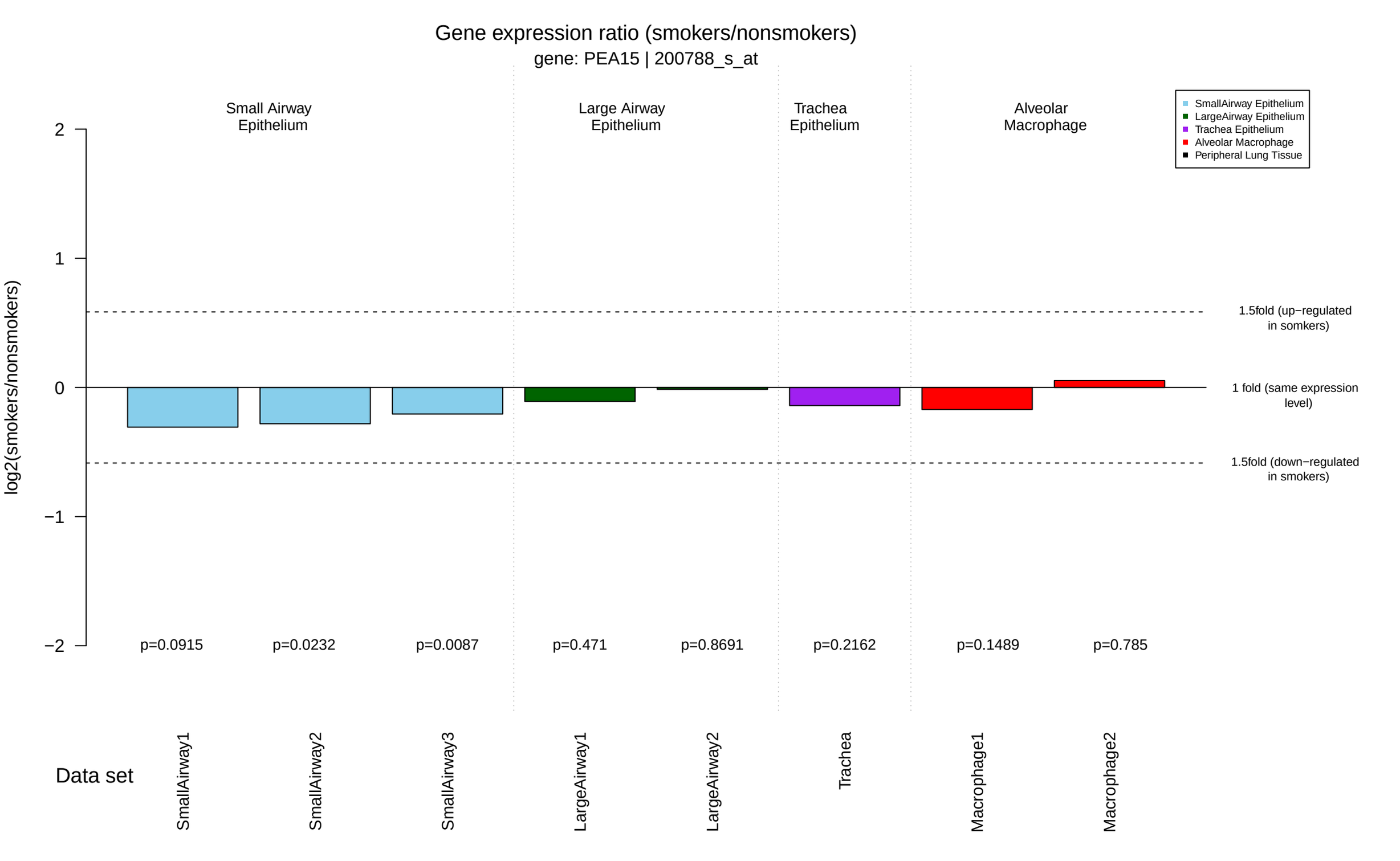
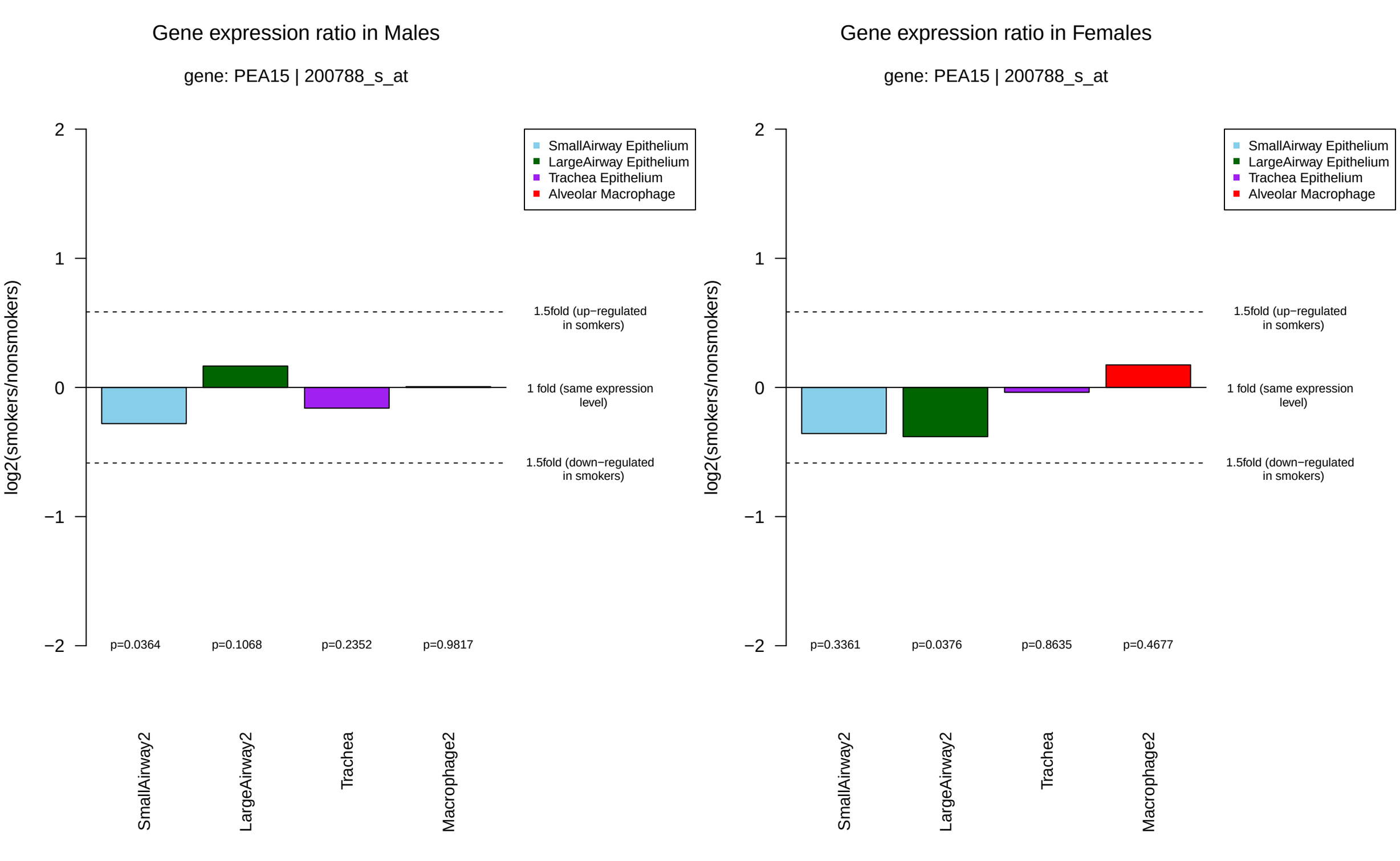
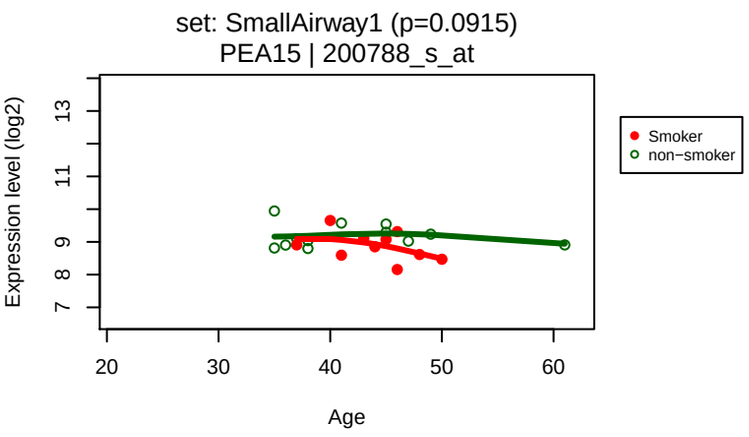
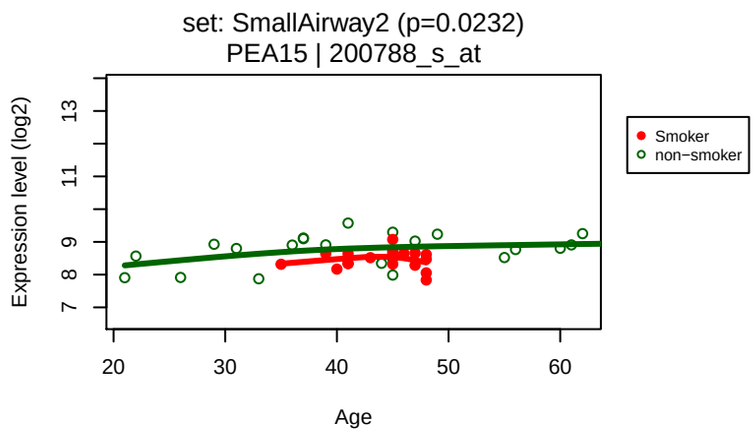
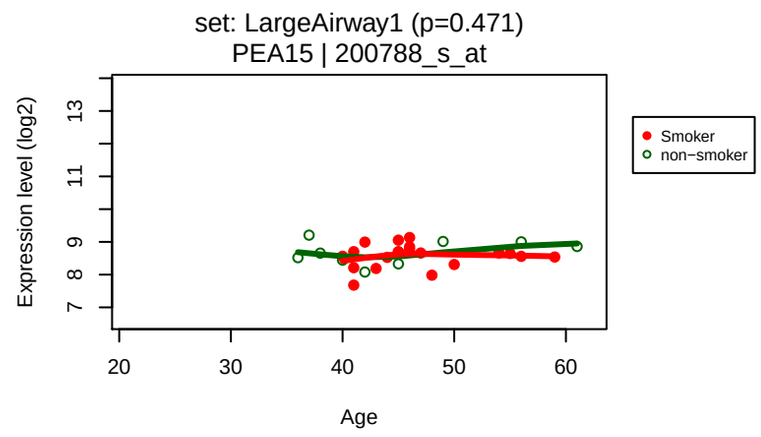

Gene expression of smokers and nonsmokers
gene: PEA15 | 200788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0915
p=0.0232
p=0.0087
p=0.471
p=0.8691
p=0.2162
p=0.1489
p=0.785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

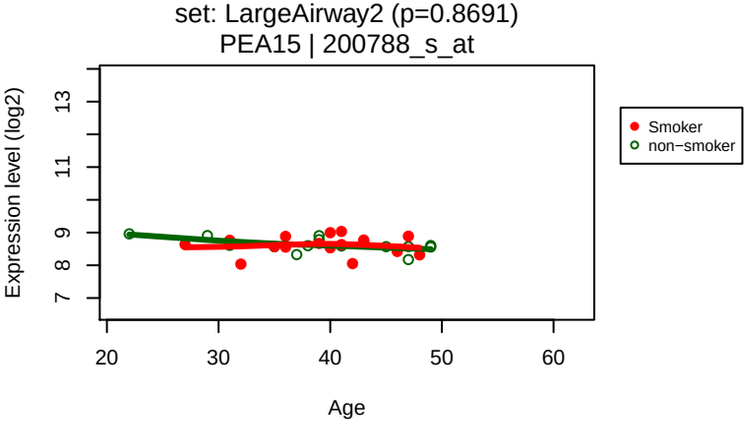

Gene expression ratio (smokers/nonsmokers)
gene: PEA15 | 200788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0915
p=0.0232
p=0.0087
p=0.471
p=0.8691
p=0.2162
p=0.1489
p=0.785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

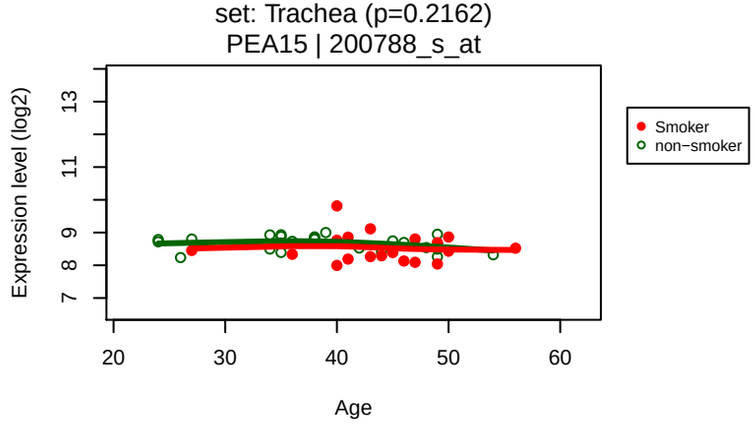

Gene expression ratio in Males
gene: PEA15 | 200788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0364
p=0.1068
p=0.2352
p=0.9817
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEA15 | 200788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3361
p=0.0376
p=0.8635
p=0.4677
SmallAirway2
LargeAirway2
Trachea
Macrophage2

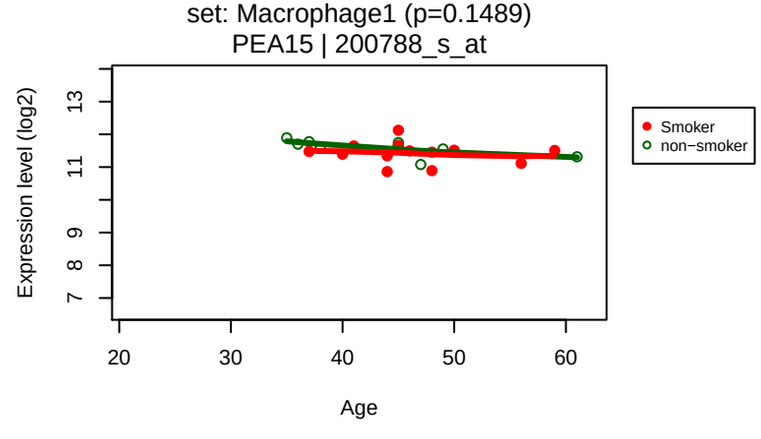

set: SmallAirway1 (p=0.0915)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker

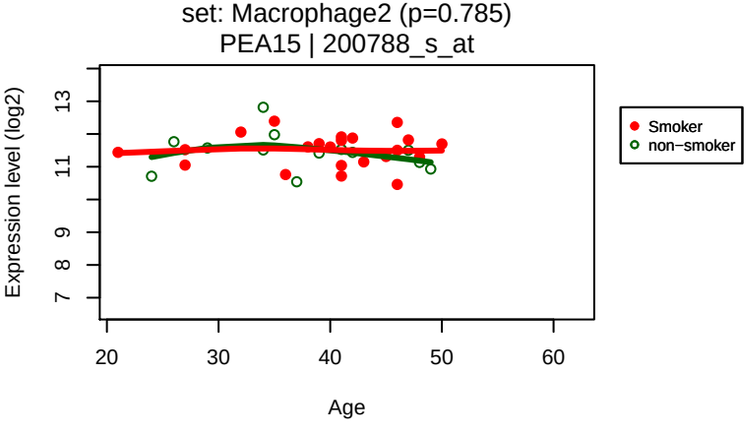

set: SmallAirway2 (p=0.0232)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker

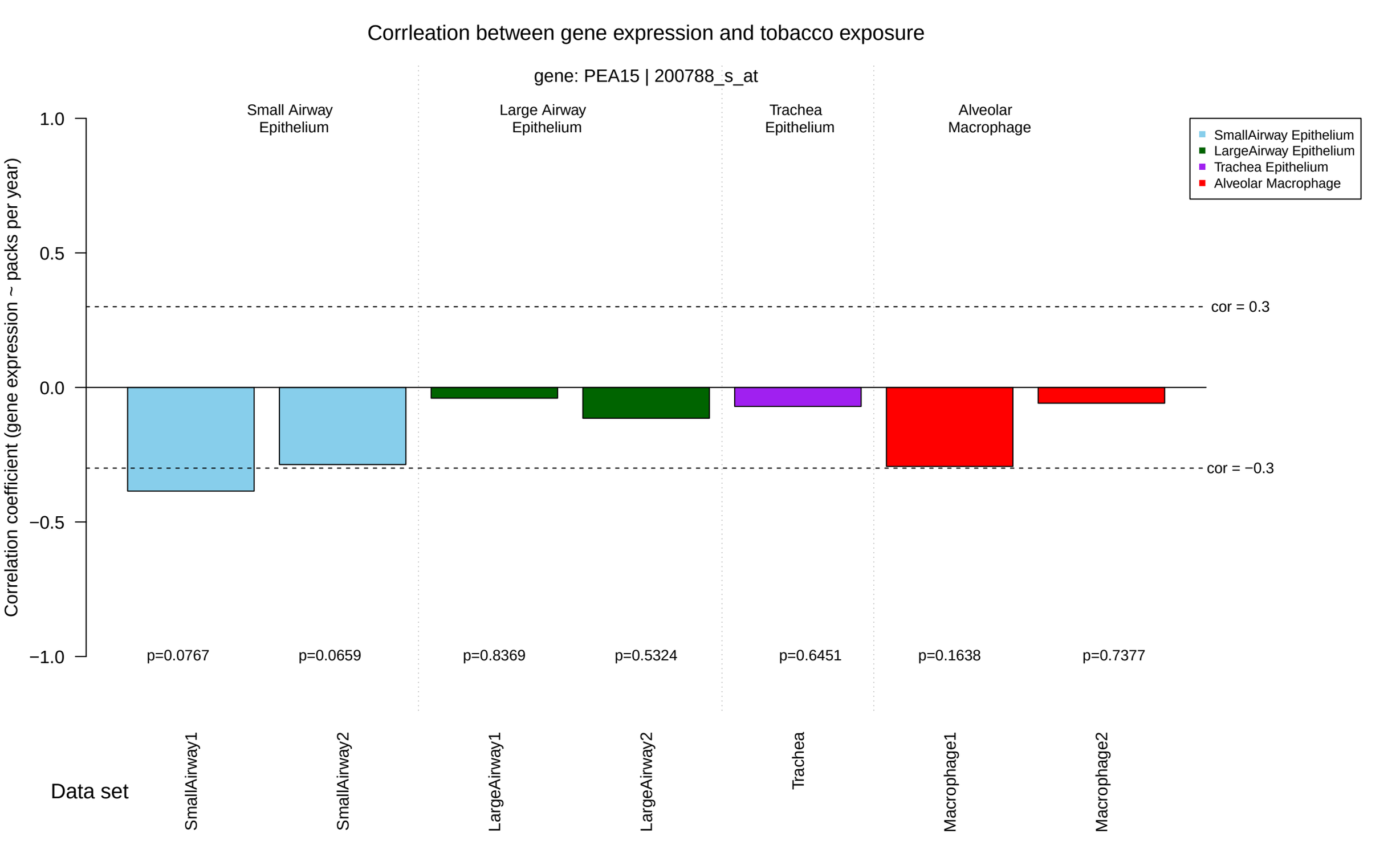

set: LargeAirway1 (p=0.471)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8691)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2162)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1489)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.785)
PEA15 | 200788_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEA15 | 200788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0767
p=0.0659
p=0.8369
p=0.5324
p=0.6451
p=0.1638
p=0.7377
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2